**SUPERSTORE SALES & PROFITABILITY ANALYSIS**

(By using m.s excel)

Submitted by

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## 1.) Project Overview: Superstore Sales & Profitability Analysis in Excel

Objective

The goal of this project is to practice the full data analysis workflow in Microsoft Excel: cleaning a raw sales dataset, analysing it with formulas and pivot tables, and presenting the results in an interactive dashboard.

Dataset

The dataset is the Superstore sales data in CSV format, with 21 columns covering orders, shipping, customers, products, and financials (sales, quantity, discount, profit). After cleaning, it contains 9,994 unique records across 4 regions and 3 product categories.

Tools and Techniques

Microsoft Excel, using Remove Duplicates, Format Cells, text functions, Excel Tables, conditional formatting, freeze panes, filters, statistical and logical formulas, pivot tables, pivot charts, slicers, and KPI cards.

Project Workflow:

Data understanding: reviewed the domain, columns, data types, and quality of the dataset.

Data cleaning: checked for missing values (none found), removed 7 duplicates, converted date and sales columns to correct data types, and standardized text with TRIM, CLEAN, PROPER, UPPER, LOWER, LEFT, MID, RIGHT, and CONCAT.

Data transformation: converted the data to an Excel Table with a table style, conditional formatting, frozen panes, and filters.

Analysis: calculated key figures with SUM, AVERAGE, MIN, MAX, COUNT, and IF, and built pivot tables by region and category.

Visualization: created four pivot charts, a Ship Mode slicer, three KPI cards, and a combined dashboard.

Key Results

Total sales: $22,97,200.86

Total profit: $2,86,397.02 (profit margin of about 12.5%)

Average sale amount: $229.86

Total quantity sold: 37,873 units

Key Insights

West is the strongest region, with the highest sales and profit.

Central has the weakest profit margin, at about 7.9%.

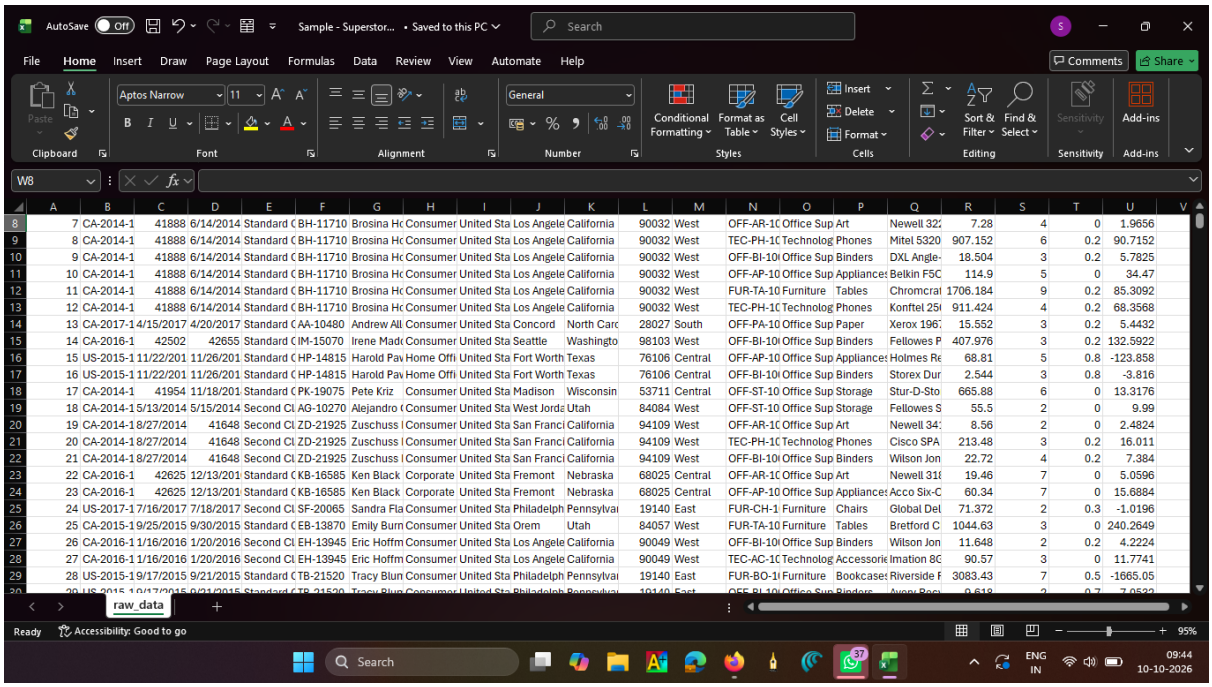
Technology earns the most revenue, while Office Supplies sells the most units.

Furniture has a very low margin of about 2.5% despite strong sales.

Outcome

The project turned raw data into a clean, structured dataset and an interactive dashboard that shows business performance briefly, and it demonstrates practical skills in data cleaning, analysis, and visualization.

Raw data before cleaning:



## 2,) Data understanding:

**Domain:** sales

**File name:** superstore sales

**File format**: csv

**Dataset description:** sales info of a superstore

**Duplicate value**: 7 duplicate values found in this dataset

**Total column's:** 21

**Total row's**: 9995

**Data type:** string, integer, float

**Column name:** row id, order id, order update, ship date, ship mode, customer id, customer name, segment, country, city, state, postal code, region, product id, category, subcategory, product name, sales, quantity, discount, profit.

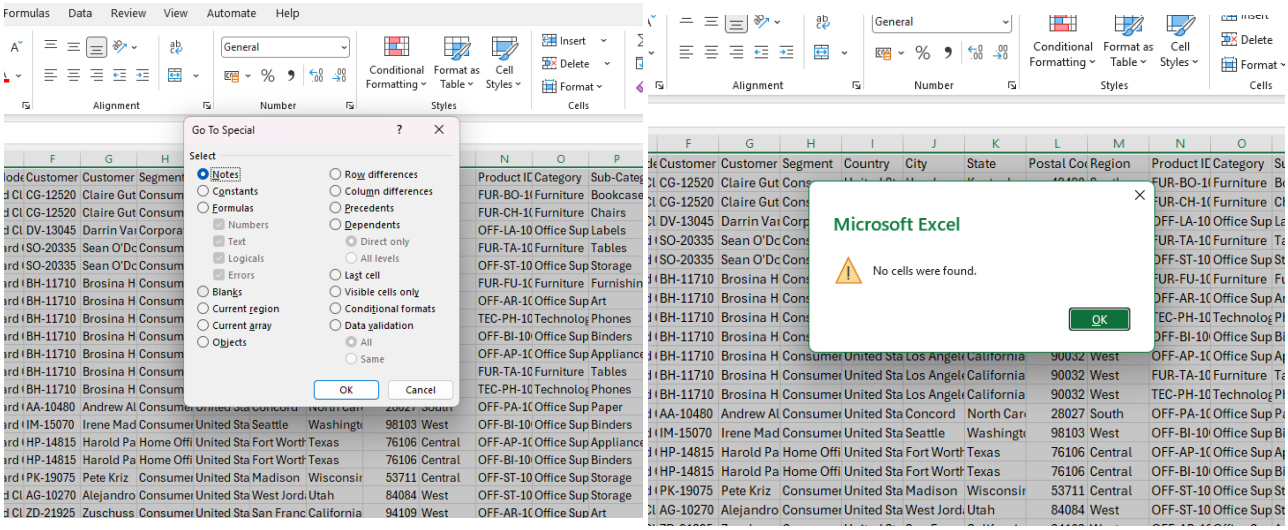
## 3.) Data Cleaning Process:

Raw data is rarely ready for analysis. It often contains duplicates, missing values, inconsistent formatting, and incorrect data types that can lead to misleading results. Data cleaning is the process of finding and fixing these issues so the dataset is accurate, consistent, and reliable.

In this project, I cleaned the dataset in Microsoft Excel. The steps included removing duplicates, handling blank or missing values, standardizing text and date formats, and correcting data types. Each step is documented below with screenshots to show the process from the raw data to the final cleaned dataset, which is ready for analysis and reporting.

## 3.1) Handling missing value:

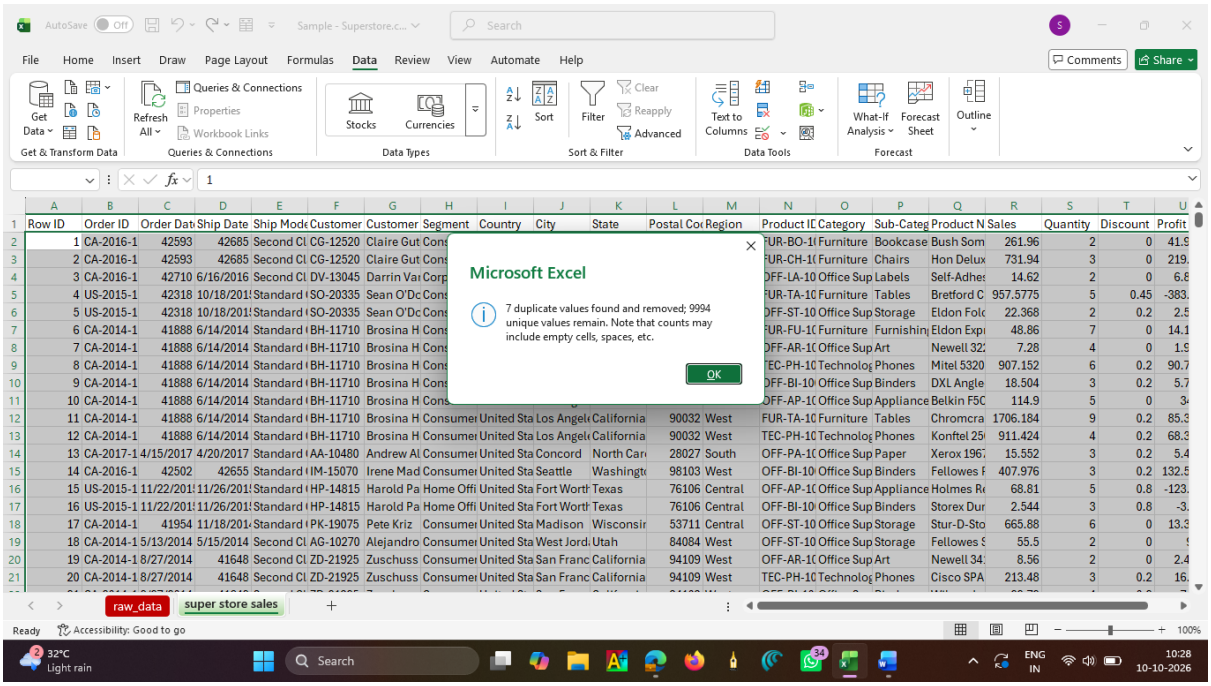
In this project, I checked the dataset for missing values using Excel tools such as filters, conditional formatting, and the COUNTBLANK function. The check confirmed that the dataset has no missing values, so no filling or removal was needed and the data stayed complete and consistent.



## 3.2) Duplicate records:

In this project, I checked the dataset for duplicates using Excel's Remove Duplicates feature. The check found 7 duplicate records, which were removed. After cleaning, the dataset contains 9,994 unique records, ready for analysis.

Duplicate finding process: I selected the whole dataset by using ctrl + a and went to data tab and the selected the remove duplicates option which is located in the data tools.

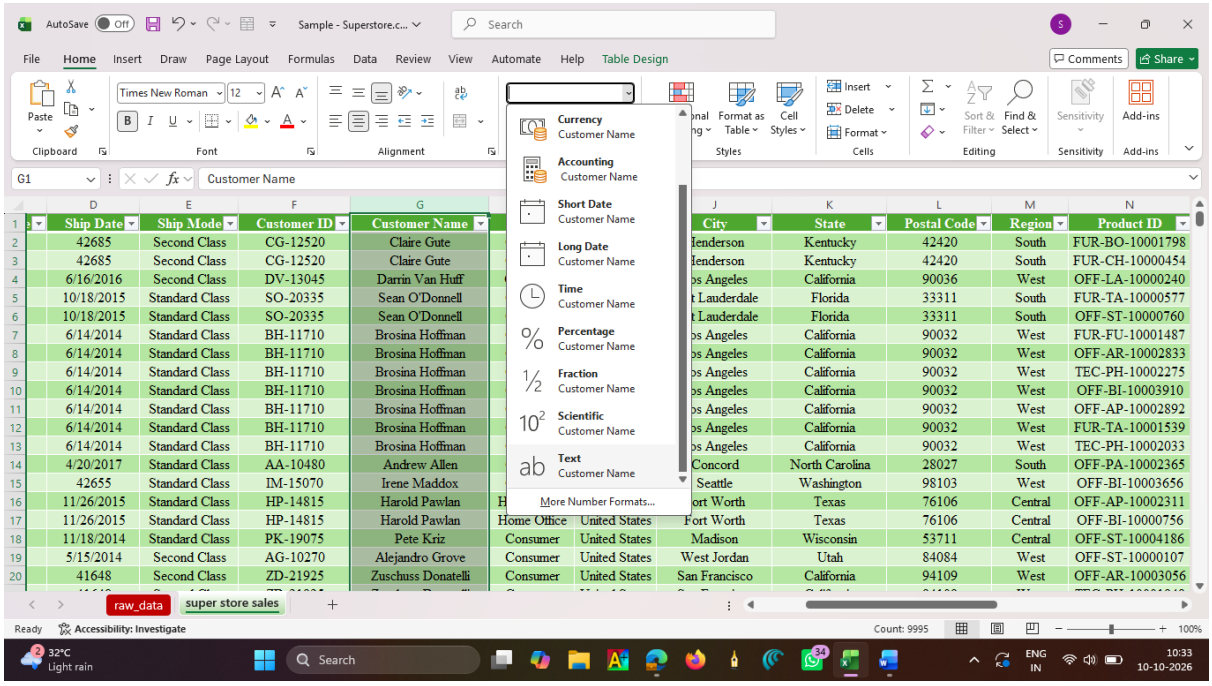



## 3.3) Data Type Conversion:

Data type conversion means making sure each column in a dataset is stored in the correct format, such as numbers, dates, or text. When a column has the wrong data type, for example dates stored as plain text or amounts stored as text, sorting, filtering, calculations, and charts can give wrong results.

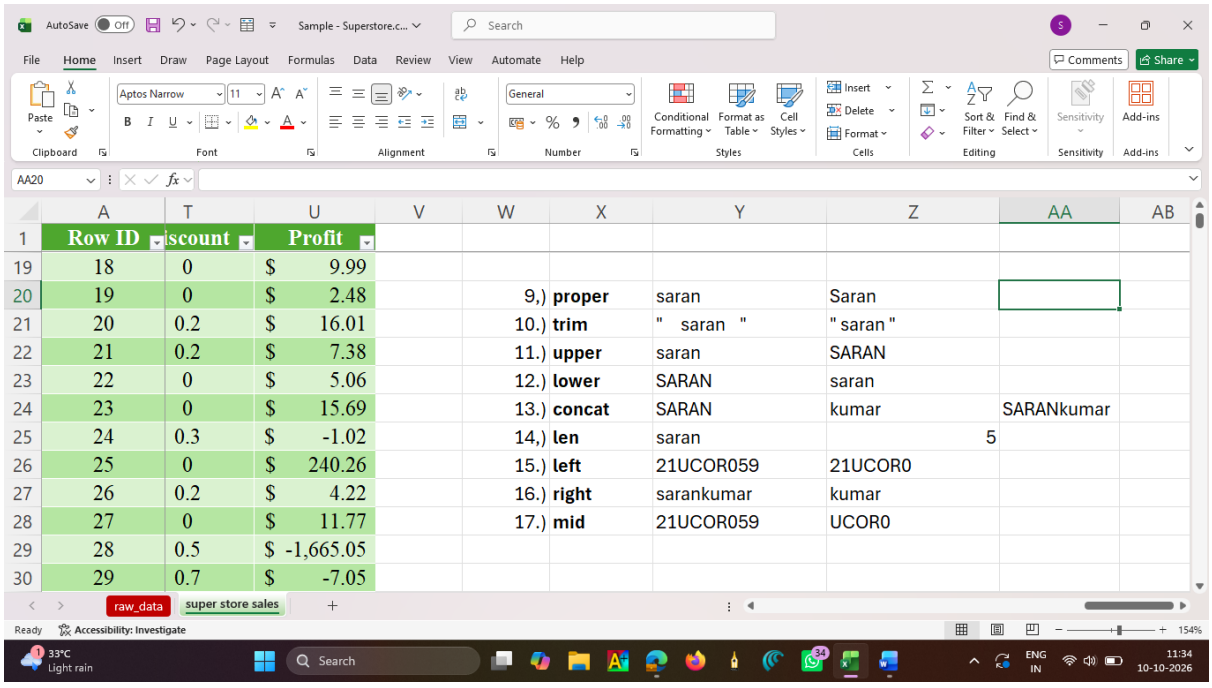
In this project, I converted the **Date** column to a proper date format and the **Sale Amount** column to a numeric format using Excel's Format Cells option. This lets Excel treat dates as real dates and amounts as numbers, so time-based analysis and calculations such as totals and averages work correctly.

(DATA TYPE CONVERSION)




## 3.4) Standardization:

In this project, I standardized the text data using Excel's text functions. TRIM removed extra spaces, CLEAN removed non-printing characters, and PROPER, UPPER, and LOWER made the letter case consistent. LEFT, MID, and RIGHT extracted specific parts of text values, and CONCAT combined values into a single field. Together these made the dataset uniform, clean, and ready for analysis.



| Row ID | Discount | Profit | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 18 | 0 | $ 9.99 | | | | | | |
| 19 | 0 | $ 2.48 | 9,) | proper | saran | Saran | | |
| 20 | 0.2 | $ 16.01 | 10.) | trim | " saran " | " saran " | | |
| 21 | 0.2 | $ 7.38 | 11.) | upper | saran | SARAN | | |
| 22 | 0 | $ 5.06 | 12.) | lower | SARAN | saran | | |
| 23 | 0 | $ 15.69 | 13.) | concat | SARAN | kumar | SARANkumar | |
| 24 | 0.3 | $ -1.02 | 14,) | len | saran | 5 | | |
| 25 | 0 | $ 240.26 | 15.) | left | 21UCOR059 | 21UCOR0 | | |
| 26 | 0.2 | $ 4.22 | 16.) | right | sarankumar | kumar | | |
| 27 | 0 | $ 11.77 | 17.) | mid | 21UCOR059 | UCOR0 | | |
| 28 | 0.5 | $ -1,665.05 | | | | | | |
| 29 | 0.7 | $ -7.05 | | | | | | |

## 4.) Data Transformation

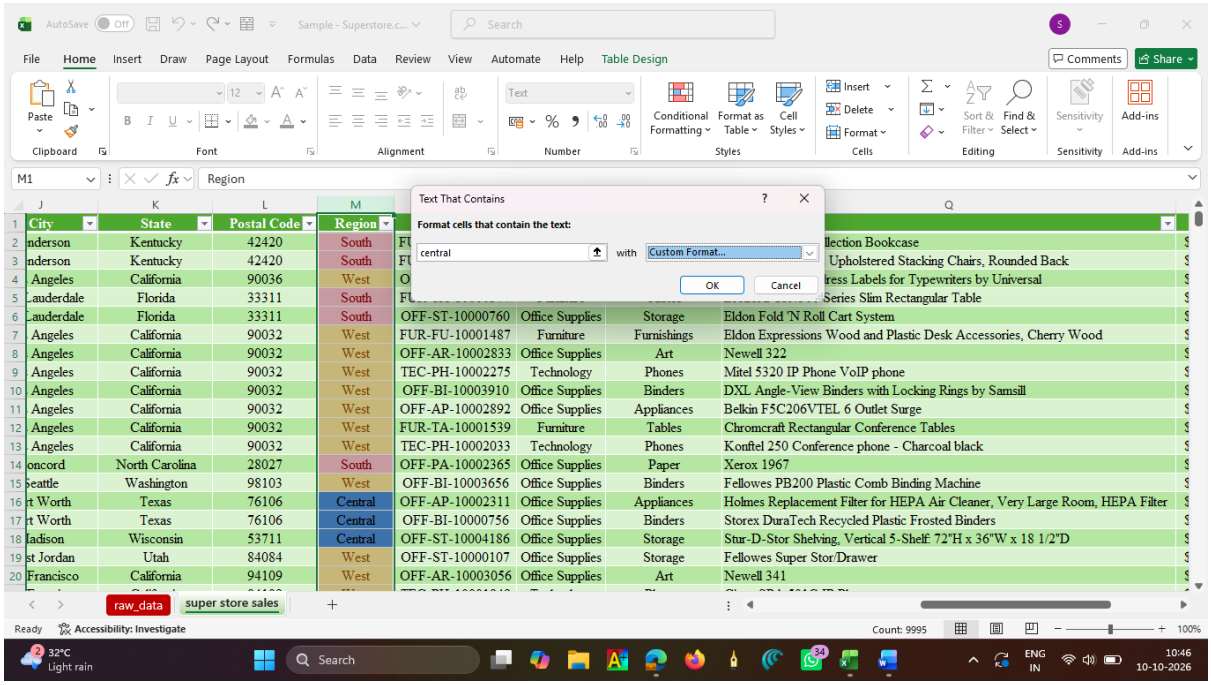
Data transformation means restructuring and formatting a cleaned dataset so it is easier to read, navigate, and analyse. Even accurate data can be hard to work with if it is presented as a plain range of cells.

In this project, I converted the dataset into an Excel Table, which makes the data structured and easier to manage, and applied a table style to improve readability. I used conditional formatting to highlight important values and patterns, froze panes to keep the header row visible while scrolling, and enabled filters (Ctrl + Shift + L) so the data can be sorted and filtered quickly. These steps turned the cleaned data into an organized, analysis-ready dataset.

## 4.1) Conditional Formatting

Conditional formatting automatically changes how cells look based on the values they contain. It makes patterns, trends, and important values stand out at a glance, without scanning the data row by row.

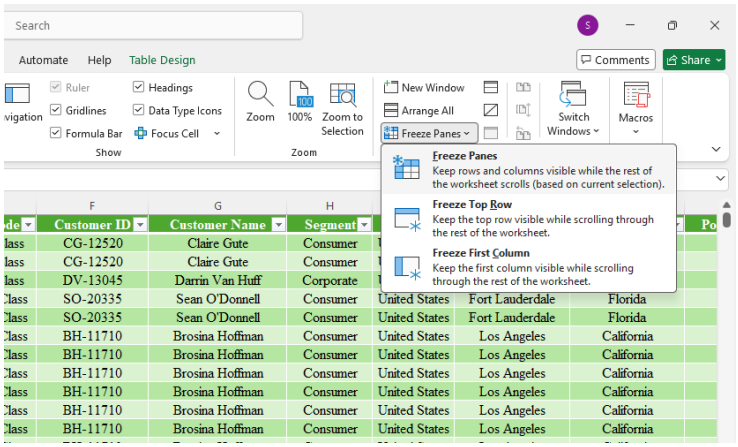
In this project, I applied conditional formatting to the table to highlight key values and make the data easier to interpret. Cells that meet specific conditions are shown in different colours or styles, which helps with spotting high and low values, identifying outliers, and reading the dataset faster.



## 4.2) Freeze Panes

Freeze Panes keeps selected rows or columns visible while you scroll through a large dataset. Without it, the headers and identifying columns disappear as you move down or across the sheet, which makes the data hard to read and easy to misinterpret.

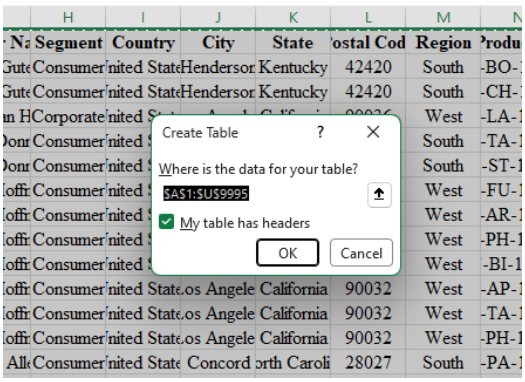
In this project, I used the Freeze Top Row and Freeze First Column options. The header row and the first column stay in place while scrolling, so every value can be matched to its column name and its record, even in a large dataset.




## 4.3) Table Creation and Table Design

An Excel Table turns a plain range of cells into a structured, manageable dataset. Tables automatically expand when new data is added, keep headers visible, and make sorting, filtering, and referencing data easier and less error-prone.

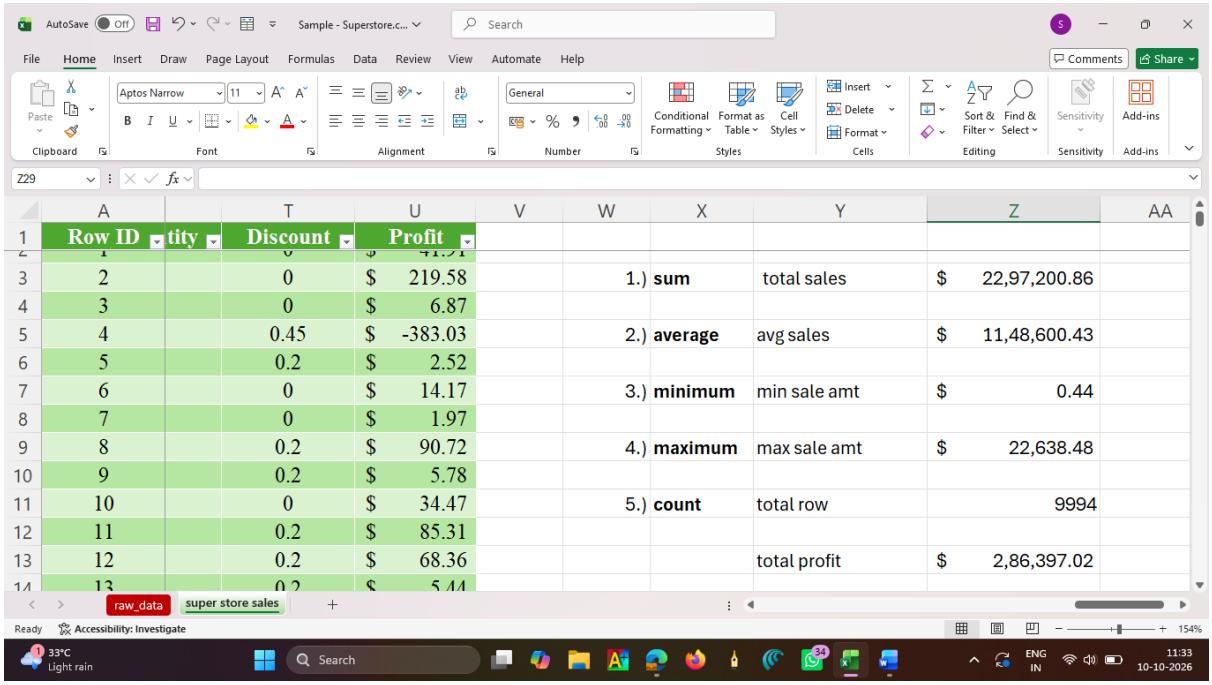
In this project, I converted the cleaned dataset into an Excel Table using Ctrl + T. I then applied a table style from the Table Design tab to improve readability, with banded rows and a distinct header style, which makes the data easier to scan and look more professional.



## 5.) Formulas & functions:

Formulas and functions are what turn raw data in Excel into useful information. A function is a built-in command that performs a calculation or manipulates data, which saves time and reduces manual errors.

In this project, I applied a range of Excel functions to the Superstore sales dataset. Statistical functions (SUM, AVERAGE, MIN, MAX, COUNT) summarized the key sales and profit figures, such as total sales, average sales, the smallest and largest sale amounts, and the total number of records. The IF function tested a logical condition, and the DATE and NOW functions returned the current date and time. Text functions (PROPER, TRIM, UPPER, LOWER, CONCAT, LEN, LEFT, RIGHT, MID) were used to clean, format, measure, and extract parts of text values. Each function is shown below with a screenshot of its result.



| Row ID | tity | Discount | Profit | | | | |
|---|---|---|---|---|---|---|---|
| 2 | | 0 | $ 219.58 | 1.) | **sum** | total sales | $ 22,97,200.86 |
| 3 | | 0 | $ 6.87 | | | | |
| 4 | | 0.45 | $ -383.03 | 2.) | **average** | avg sales | $ 11,48,600.43 |
| 5 | | 0.2 | $ 2.52 | | | | |
| 6 | | 0 | $ 14.17 | 3.) | **minimum** | min sale amt | $ 0.44 |
| 7 | | 0 | $ 1.97 | | | | |
| 8 | | 0.2 | $ 90.72 | 4.) | **maximum** | max sale amt | $ 22,638.48 |
| 9 | | 0.2 | $ 5.78 | | | | |
| 10 | | 0 | $ 34.47 | 5.) | **count** | total row | 9994 |
| 11 | | 0.2 | $ 85.31 | | | | |
| 12 | | 0.2 | $ 68.36 | | | total profit | $ 2,86,397.02 |

(In this excel sheet I used sum, average, minimum, maximum,count by using formulas)

Common built functions:

SUM Function

The SUM function adds all the numbers in a range. In this project, it was used to calculate the total sales (and total profit) of the dataset, which came to $22,97,200.86 in total sales.

Step: Click the result cell, type =SUM(Table1[Sales]), and press Enter.

AVERAGE Function

The AVERAGE function returns the mean of a set of numbers. In this project, it was used to find the average sale amount per record, which shows the typical value of a sale.

Step: Click the result cell, type =AVERAGE(Table1[Sales]), and press Enter.

MIN Function

The MIN function returns the smallest value in a range. In this project, it was used to find the lowest sale amount in the dataset, which is $0.44.

Step: Click the result cell, type =MIN(Table1[Sales]), and press Enter.

MAX Function

The MAX function returns the largest value in a range. In this project, it was used to find the highest sale amount in the dataset, which is $22,638.48.

Step: Click the result cell, type =MAX(Table1[Sales]), and press Enter.

COUNT Function

The COUNT function counts the cells in a range that contain numbers. In this project, it was used to find the total number of records in the dataset, which is 9,994.

Step: Click the result cell, type =COUNT(Table1[Row ID]), and press Enter.

IF Function

The IF function tests a condition and returns one result if it is true and another if it is false. In this project, it was used to check a logical condition on the data, and it returned “true” for the cell tested.

Step: Click the result cell, type =IF(logical_test, "true", "false"), for example =IF(Z3>1000,"true","false"), and press Enter.

DATE Function

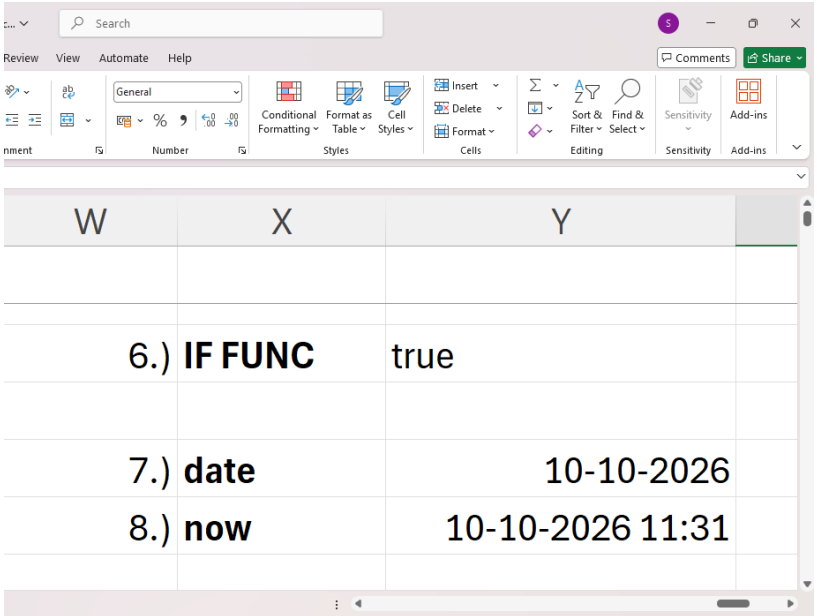
The DATE function builds a date from a year, month, and day that you supply. In this project, it was used to create a proper date value, which Excel displays as 10-10-2026 and can use in date calculations.

Step: Click the result cell, type =DATE(2026,10,10), and press Enter.

NOW Function

The NOW function returns the current date and time, and it updates every time the sheet recalculates. In this project, it was used to show the current date and time, displayed as 10-10-2026 11:31.

Step: Click the result cell, type =NOW(), and press Enter.



TRIM Function

The TRIM function removes extra spaces from text, leaving only single spaces between words. In this project, it was used to clean text with extra spaces around it, turning “ saran “ into “saran”.

Step: Click the result cell, type =TRIM(Y21), and press Enter.

PROPER Function

The PROPER function capitalizes the first letter of each word and makes the rest lowercase. In this project, it was used to standardize text, converting “saran” into “Saran”.

Step: Click the result cell, type =PROPER(Y20), and press Enter.

UPPER Function

The UPPER function converts all letters in a text value to capital letters. In this project, it was used to convert “saran” into “SARAN”.

Step: Click the result cell, type =UPPER(Y22), and press Enter.

LOWER Function

The LOWER function converts all letters in a text value to lowercase. In this project, it was used to convert “SARAN” into “saran”.

Step: Click the result cell, type =LOWER(Y23), and press Enter.

CONCAT Function

The CONCAT function joins text from two or more cells into one. In this project, it was used to combine “SARAN” and “kumar” into “SARANkumar”.

Step: Click the result cell, type =CONCAT(Y24,Z24), and press Enter.

LEN Function

The LEN function counts the number of characters in a text value, including spaces. In this project, it was used to count the characters in “saran”, which returned 5.

Step: Click the result cell, type =LEN(Y25), and press Enter.

LEFT Function

The LEFT function extracts a chosen number of characters from the start of a text value. In this project, it was used to extract the first 7 characters of “21UCOR059”, which returned “21UCOR0”.

Step: Click the result cell, type =LEFT(Y26,7), and press Enter.

RIGHT Function

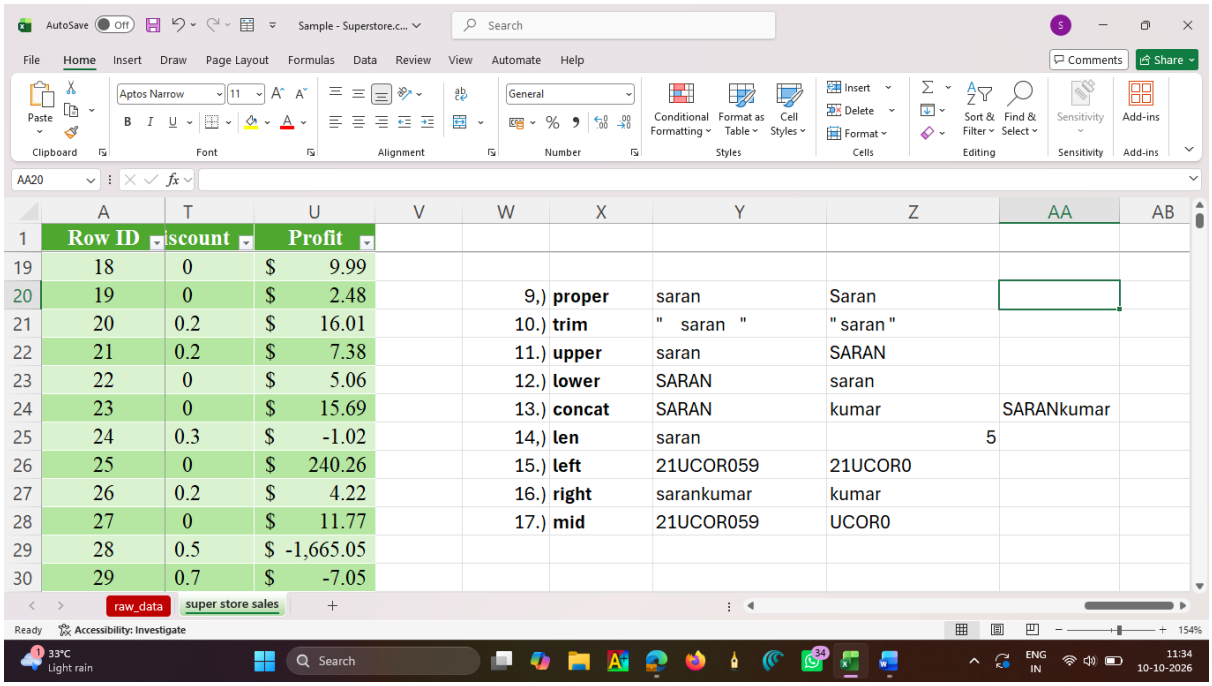
The RIGHT function extracts a chosen number of characters from the end of a text value. In this project, it was used to extract the last 5 characters of “sarankumar”, which returned “kumar”.

Step: Click the result cell, type =RIGHT(Y27,5), and press Enter.

MID Function

The MID function extracts characters from the middle of a text value, starting at a position you choose. In this project, it was used to extract 5 characters starting from the 3rd position of “21UCOR059”, which returned “UCOR0”.

Step: Click the result cell, type =MID(Y28,3,5), and press Enter.




| | Row ID | Discount | Profit | | | | | |
|---|---|---|---|---|---|---|---|---|
| 19 | 18 | 0 | $ 9.99 | | | | | |
| 20 | 19 | 0 | $ 2.48 | 9,) | proper | saran | Saran | |
| 21 | 20 | 0.2 | $ 16.01 | 10.) | trim | " saran " | " saran " | |
| 22 | 21 | 0.2 | $ 7.38 | 11.) | upper | saran | SARAN | |
| 23 | 22 | 0 | $ 5.06 | 12.) | lower | SARAN | saran | |
| 24 | 23 | 0 | $ 15.69 | 13.) | concat | SARAN | kumar | SARANkumar |
| 25 | 24 | 0.3 | $ -1.02 | 14,) | len | saran | 5 | |
| 26 | 25 | 0 | $ 240.26 | 15.) | left | 21UCOR059 | 21UCOR0 | |
| 27 | 26 | 0.2 | $ 4.22 | 16.) | right | sarankumar | kumar | |
| 28 | 27 | 0 | $ 11.77 | 17.) | mid | 21UCOR059 | UCOR0 | |
| 29 | 28 | 0.5 | $ -1,665.05 | | | | | |
| 30 | 29 | 0.7 | $ -7.05 | | | | | |

## 6.)Pivot Tables:

A PivotTable is an Excel tool that summarizes a large dataset quickly without writing formulas. The Superstore dataset has 9,994 rows, so it is hard to read directly. Four pivot tables were created on the p.table & charts sheet to summarize sales and profit by sub-category, region, category and quantity. They are the data source for the four dashboard charts.

| Pivot table | Rows | Values | Used for |
|---|---|---|---|
| 1 | Sub-Category | Sum of Sales | Bar chart |
| 2 | Region | Sum of Profit | Column chart |
| 3 | Quantity | Sum of Sales | Line chart |
| 4 | Category | Sum of Sales | Pie chart |

Short Analysis

Sales by Sub-Category (total ₹/$22,97,200.86)

Phones (3,30,007.05) and Chairs (3,28,449.10) are the top two sub-categories. Together they make up about 29% of total sales.

Fasteners (3,024.28) is the lowest, followed by Labels (12,486.31) and Envelopes (16,476.40).

Profit by Region (total 2,86,397.02)

West earns the most profit (1,08,418.45), about 38% of the total.

East is second (91,522.78).

Central is the lowest (39,706.36), so it is the region with the most room to improve.

Sales by Quantity

Sales peak at a quantity of 3 (4,22,060.60), followed by 5 (4,15,661.70) and 4 (3,23,670.99).

Orders of 2 to 5 units make up about 63% of all sales.

Sales drop sharply above 9 units, which means large-quantity orders are rare.

Sales by Category

Technology leads (8,36,154.03), about 36% of sales.

Furniture (7,41,999.80) and Office Supplies (7,19,047.03) are close behind, so the three categories contribute fairly evenly.

Conclusion: The pivot tables show that sales are driven mainly by Technology products, particularly Phones, and by Furniture such as Chairs. Profit is strongest in the West region and weakest in Central. Most sales come from orders of 2 to 5 units.

| Row Labels | Sum of Sales |
|---|---|
| Accessories | $ 1,67,380.32 |
| Appliances | $ 1,07,532.16 |
| Art | $ 27,118.79 |
| Binders | $ 2,03,412.73 |
| Bookcases | $ 1,14,880.00 |
| Chairs | $ 3,28,449.10 |
| Copiers | $ 1,49,528.03 |
| Envelopes | $ 16,476.40 |
| Fasteners | $ 3,024.28 |
| Furnishings | $ 91,705.16 |
| Labels | $ 12,486.31 |
| Machines | $ 1,89,238.63 |
| Paper | $ 78,479.21 |
| Phones | $ 3,30,007.05 |
| Storage | $ 2,23,843.61 |
| Supplies | $ 46,673.54 |
| Tables | $ 2,06,965.53 |
| **Grand Total** | **$ 22,97,200.86** |

| Row Labels | Sum of Profit |
|---|---|
| Central | $ 39,706.36 |
| East | $ 91,522.78 |
| South | $ 46,749.43 |
| West | $ 1,08,418.45 |
| **Grand Total** | **$ 2,86,397.02** |

| Row Labels | Sum of Sales |
|---|---|
| Furniture | $ 7,41,999.80 |
| Office Supplies | $ 7,19,047.03 |
| Technology | $ 8,36,154.03 |
| **Grand Total** | **$ 22,97,200.86** |

| Row Labels | Sum of Sales |
|---|---|
| 1 | $ 53,251.93 |
| 2 | $ 2,89,091.48 |
| 3 | $ 4,22,060.60 |
| 4 | $ 3,23,670.99 |
| 5 | $ 4,15,661.70 |
| 6 | $ 2,07,122.32 |
| 7 | $ 2,39,908.37 |
| 8 | $ 1,17,760.18 |
| 9 | $ 1,28,505.59 |
| 10 | $ 24,056.66 |
| 11 | $ 15,772.48 |
| 12 | $ 14,868.41 |
| 13 | $ 32,338.19 |
| 14 | $ 13,131.96 |
| **Grand Total** | **$ 22,97,200.86** |

(full picture of 4 pivot table)

## 7.) Pivot charts:

Bar chart (Sales by Sub-Category):

Introduction: This bar chart shows the sales distribution across different product sub-categories, such as Tables, Storage, Paper, Labels, Copiers, and Bookcases.

Short Analysis: Copiers and Bookcases generate some of the highest sales, followed by Storage and Paper. Labels and Fasteners contribute comparatively lower sales. This indicates that customer demand varies significantly across product sub-categories.

Conclusion: The business should focus on high-performing sub-categories while exploring ways to improve sales of lower-performing products.

Column chart (Profit by Region):

Introduction: This column chart compares the profit generated by four regions: Central, East, South, and West.

Short Analysis: The West region generates the highest profit at approximately $108,418, followed by the East at $91,523. The South earns around $46,749, while the Central region generates approximately $39,706, making it the lowest among the four regions.

Conclusion: The West and East regions are the strongest profit contributors. The business should investigate opportunities to improve profitability in the Central and South regions.

Line chart (Sales by Quantity):

Introduction: This line chart displays sales values across 14 quantity-related data points, helping visualize changes and fluctuations in sales.

Short Analysis: Sales rise sharply during the early data points and reach a peak around the third and fifth points, at approximately $4.2 million and $4.15 million, respectively. Afterward, sales fluctuate and generally decline, with some smaller increases toward the end.

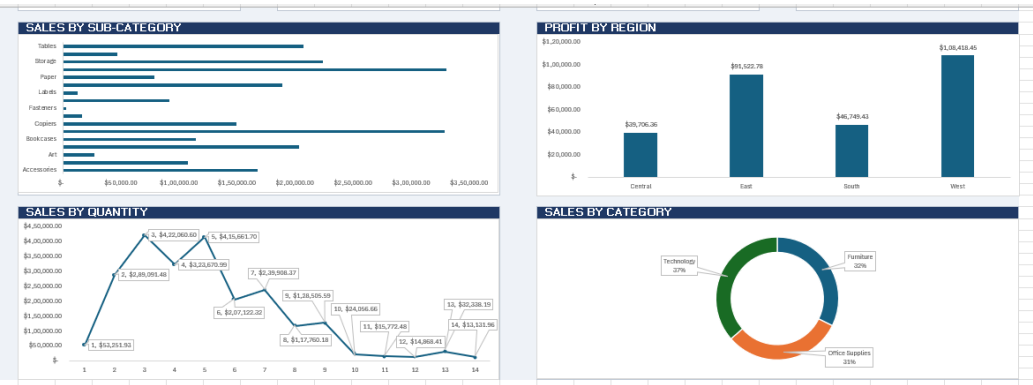
Conclusion: Sales performance shows considerable variation over the displayed data points. The business should investigate the reasons for the peaks and declines to identify opportunities for more consistent sales performance.

Sales by Category

Introduction: This doughnut chart illustrates the percentage distribution of sales across three major categories: Technology, Furniture, and Office Supplies.

Short Analysis: Technology accounts for the largest share of sales at 37%, followed by Furniture at 32% and Office Supplies at 31%. The distribution is relatively balanced, although Technology holds a slight lead.

Conclusion: All three categories contribute substantially to overall sales. The business should maintain a balanced product strategy while exploring opportunities to increase sales in the Technology category.




(Full picture of the 4 charts)

## 8.) Slicers:

Region Slicer:

The Region slicer allows users to filter the dashboard data based on four regions: Central, East, South, and West. It helps compare sales and profit performance across different geographical areas.

Category Slicer

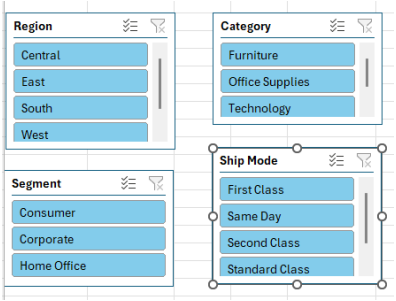
The Category slicer enables users to filter data by three major product categories: Furniture, Office Supplies, and Technology. It helps analyze the sales and performance of each product category separately.

Segment Slicer

The Segment slicer allows users to view data for three customer segments: Consumer, Corporate, and Home Office. It helps understand customer purchasing patterns and compare business performance across different customer groups.

Ship Mode Slicer

The Ship Mode slicer filters the data according to four shipping methods: First Class, Same Day, Second Class, and Standard Class. It helps analyze sales performance based on shipping preferences and delivery methods



## 9.) KPI Cards:

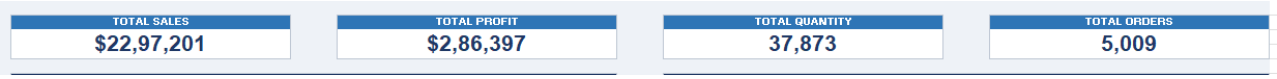
KPI (Key Performance Indicator) cards provide a quick overview of the overall business performance by displaying four important metrics: Total Sales, Total Profit, Total Quantity, and Total Orders.

Total Sales: Displays the total revenue generated by the business, amounting to $22,97,201. It helps evaluate overall sales performance.

Total Profit: Shows the total profit earned by the business, amounting to $2,86,397. It helps measure the business's profitability.

Total Quantity: Represents the total number of product units sold, amounting to 37,873. It helps assess product demand and sales volume.

Total Orders: Indicates the total number of orders placed, amounting to 5,009. It helps understand overall customer purchasing activity.




## 10.) Dashboard:

The Superstore Sales Dashboard is an interactive Excel dashboard designed to analyze overall business performance. It provides key performance indicators (KPIs), including Total Sales, Total Profit, Total Quantity, and Total Orders. It also includes four charts to visualize sales by sub-category, profit by region, sales by quantity, and sales by category. The slicers allow users to filter the data by segment, region, ship mode, and category for more detailed analysis.

Short Analysis of the dashboard:

Overall Performance: The business generated total sales of $2,297,201, a total profit of $286,397, sold 37,873 units, and received 5,009 orders.

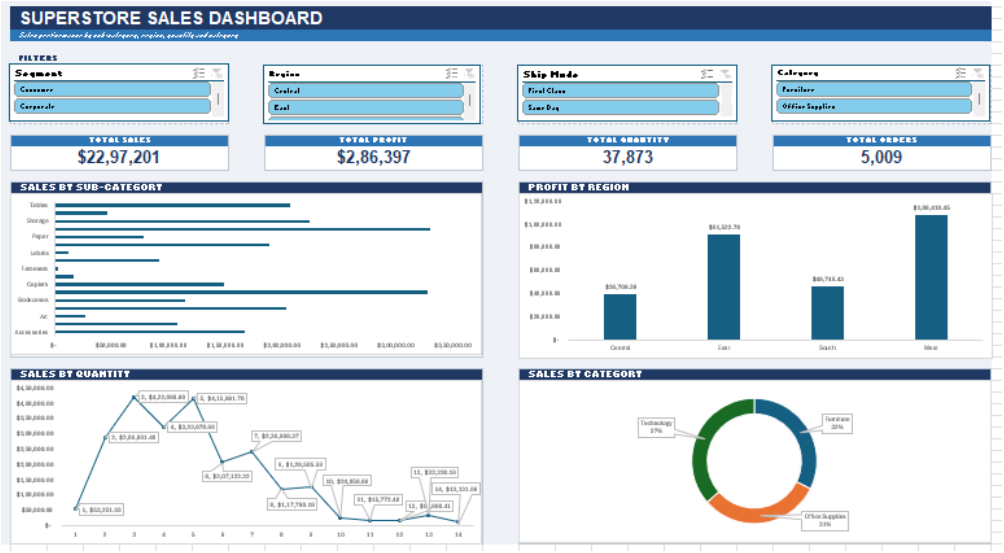
Sales by Sub-Category: Copiers and Bookcases are among the highest-performing sub-categories, while some others contribute relatively lower sales.

Profit by Region: The West region generates the highest profit, followed by the East, while the Central region records the lowest profit.

Sales by Quantity: The chart shows fluctuations across the data points, with higher values in the earlier points and a general decline afterward.

Sales by Category: Technology leads with 37% of sales, followed by Furniture at 32% and Office Supplies at 31%.

Interactive Filters: The slicers make it easy to compare performance across different customer segments, regions, shipping methods, and product categories.



(Full view of the superstore sales dashboard)

## 11.) Insights Report:

Based on the cleaned Superstore dataset (9,994 records, orders from 2014 to 2017). All figures are in USD.

**1. What happened?**

• The business generated $2.30M in sales and $286K in profit, a profit margin of 12.5%. It sold 37,873 units across 5,009 orders, with an average sale of $229.86.

• Sales are healthy, but profit is uneven. About 18.7% of all order lines (1,871) lose money, and those losses add up to roughly $156K, which cuts heavily into total profit.

**2. Why did it happen?**

• Discounting is the main cause. Lines sold with no discount earn a 29.5% margin. A 0 to 20% discount earns 11.9%, a 20 to 40% discount loses 15.3%, and a discount above 40% loses 77.4%.

• Lines discounted by 30% or more account for about 89% of all losses.

• Furniture is discounted more heavily (average 17.4%) than Technology (13.2%), and several furniture items sell below cost.

• The Central region gives the highest average discount at 24%, compared with 10.9% in the West. This is the main reason Central has the weakest margin.

**3. Where did it happen?**

• West is the strongest region: 31.6% of sales, 37.9% of profit and a 14.9% margin. East follows with a 13.5% margin and South with 11.9%.

• Central is the weakest: it brings in 21.8% of sales but only 13.9% of profit, a margin of just 7.9%. Furniture in Central loses money (margin of -1.8%).

• By state, California ($76K profit) and New York ($74K) lead. Texas (-$25.7K), Ohio (-$17.0K) and Pennsylvania (-$15.6K) lose money, even though Texas is the third-largest state by sales.

**4. Who or what contributed most?**

• Category: Technology contributes the most sales ($836K, 36.4%) and the most profit ($145K). Office Supplies adds $122K profit, while Furniture adds only $18K on $742K of sales.

• Sub-category: Phones ($330K) and Chairs ($328K) lead sales. Copiers lead profit ($55.6K, a 37.2% margin), followed by Accessories ($41.9K) and Paper ($34.1K).

• Segment: Consumer customers bring in about half of all sales ($1.16M), while Home Office has the best margin (14.0%).

• Region: West contributes the most to both sales and profit.

• Shipping: Standard Class carries about 59% of sales.

**5. What changed over time?**

• Sales: $484K in 2014, $471K in 2015 (down 2.8%), $609K in 2016 (up 29.5%) and $733K in 2017 (up 20.4%). Sales grew about 51% from 2014 to 2017.

• Profit: $49.5K, $61.6K, $81.8K and $93.4K, which is growth of about 89% over the same period.

• Margin: improved from 10.2% in 2014 to about 13% in 2015 and 2016, and held at 12.7% in 2017. Growth in sales did not come at the cost of margin.

• Note: the Order Date column had mixed formats in the raw file, so the monthly figures in this report were calculated after correcting the dates.

**6. What is performing poorly?**

• Tables lose $17.7K (margin of -8.6%), Bookcases lose $3.5K (-3.0%) and Supplies lose $1.2K (-2.5%).

• Machines earn only a 1.8% margin on $189K of sales, and Chairs earn 8.1% on $328K.

• Furniture as a whole has a 2.5% margin despite 32% of total sales.

• Central region (7.9% margin) and the states of Texas, Ohio and Pennsylvania are loss areas.

• Small sub-categories such as Fasteners, Labels and Envelopes have low sales ($3K to $16K), but their margins are high (31% to 44%), so they are not a profit problem.

**7. What are the major trends?**

• Growth: strong sales growth in 2016 and 2017 with a steady margin of about 13%.

• Seasonality: September to December brings in about 52% of yearly sales. November (15.3%) and December (14.2%) are the peaks, and January and February are the weakest months (4.1% and 2.6%).

• Order size: orders of 2 to 5 units make up about 63% of sales, and sales drop sharply above 9 units.

• Product mix: Office Supplies sells the most units (22,906 of 37,873) but only 31% of sales, while Technology sells just 6,939 units and earns the most revenue.

**8. Are there anomalies?**

• The largest sale ($22,638.48, a Cisco TelePresence system in the South) actually lost $1,811 because of a 50% discount.

• The largest loss is -$6,599.98, on a Cubify 3D printer sold at a 70% discount.

• 933 order lines carry a discount of 40% or more, and together they lose about $99.6K.

• Sale values are heavily skewed: the average is $229.86, but the largest is $22,638.48 and the smallest is $0.44. A few large orders can swing the totals.

• Data quality: 7 duplicates were removed and the date column had mixed formats.

**9. What should management do?**

• Control discounts. Require approval for any discount above 20%, and stop discounts of 40% or more, which lose money on almost every sale.

• Review Tables, Bookcases and Supplies. Raise prices, renegotiate supplier costs or reduce the range.

• Fix the Central region. Review its discount policy and its Furniture sales, and investigate Texas, Ohio and Pennsylvania.

• Look at Machines. A 1.8% margin on $189K of sales needs a pricing or cost review.

• Protect the strengths. Keep investing in Technology (Copiers, Accessories, Phones), Paper and the West region.

**10. What future action is recommended?**

• Next quarter: put the discount approval rule in place, finish the Tables and Bookcases price review and audit the Central region.

• Seasonal planning: build inventory and marketing plans around the September to December peak.

• Targets: aim to lift Central's margin toward the company average of 12.5% and bring loss-making sub-categories to break-even.

• Dashboard upgrades: add a Profit Margin % KPI card, a Year slicer, and a chart of discount against profit.

## Conclusion:

This project took the raw Superstore sales dataset through the full data analysis workflow in Microsoft Excel. The dataset had 9,995 rows and 21 columns. It was cleaned by checking for missing values, removing 7 duplicate records and correcting data types, and standardized with text functions. It was then organized as an Excel Table with conditional formatting, frozen panes and filters. After cleaning, 9,994 unique records remained and were ready for analysis.

Formulas such as SUM, AVERAGE, MIN, MAX and COUNT, together with four pivot tables, summarized the business performance. The business generated $22,97,200.86 in sales and $2,86,397.02 in profit, a margin of about 12.5%. It sold 37,873 units across 5,009 orders. The results were presented in an interactive dashboard with four pivot charts, four slicers (Segment, Region, Ship Mode and Category) and four KPI cards.

Key findings:

Technology is the largest category by sales (about 36%) and by profit, while Furniture earns only a 2.5% margin despite strong sales.

The West region performs best in both sales and profit, while the Central region has the weakest margin at about 7.9%.

Discounting is the main reason for losses. Orders with discounts above 20% lose money, and about 18.7% of all order lines are loss-making.

Sub-categories such as Tables, Bookcases and Supplies sell well but lose money, while Copiers, Paper and Accessories earn the highest margins.

Sales and profit grew from 2014 to 2017, and most sales come from orders of 2 to 5 units.

Recommendations: the business should limit heavy discounts, review the pricing of loss-making products, and improve performance in the Central region. At the same time, it should keep investing in its strongest products and regions.

Skills gained: this project gave me practical experience in data cleaning, formulas, pivot tables, charts, slicers and dashboard design in Excel. It also showed me how to turn raw data into clear insights that support business decisions.

Future scope: the dashboard can be extended with a Year slicer, a Profit Margin % KPI card, and a monthly sales trend once the Order Date column is cleaned.

One thing to check: the dashboard shows Total Orders as 5,009, but your project overview lists only three KPI cards. Make sure both sections say four KPI cards so the document is consistent.